TI BA II Plus: Basic Usage

- **Setup.** `2ND FORMAT 9 ENTER` shows 9 decimals. `2ND P/Y 1 ENTER` sets one payment per year. `2ND BGN` should show END (`2ND SET` toggles). `2ND QUIT` leaves a worksheet.
- **Arithmetic.** Type numbers and operators in order, then `=`. Use `(` and `)` to group. Example pattern: `( 1 + 0.05 ) y^x 3 =` raises 1.05 to the 3rd power. `1/x` takes the reciprocal. `+/-` changes the sign. Enter percents as decimals in arithmetic (5% is `0.05`).
- **Memory.** `STO` then a digit saves the displayed value. `RCL` then the digit brings it back. `2ND CLR WORK` clears memories.
- **Time value of money (TVM).** Keys `N` (periods), `I/Y` (rate per period, **entered in percent**: 5 for 5%), `PV`, `PMT`, `FV`. Type a value then the key to store it. Press `CPT` then a key to solve for it. Press `2ND CLR TVM` before every new problem. Cash you pay out is negative and cash you receive is positive, so enter a price as a negative `PV` (type the number, then `+/-`, then `PV`).
- **Bonds.** Price: enter `N`, `I/Y` (yield), `PMT` (coupon), `FV` (face), `CPT PV`. YTM: enter `N`, `PV` (negative price), `PMT`, `FV`, `CPT I/Y`. Semiannual coupons: $N = 2T$, $I/Y = y/2$, $PMT = C$, and double an I/Y answer for the annual rate.
- **Dates.** `2ND DATE`, type `DT1` as `m.ddyy` and `ENTER ↓`, then `DT2`. `2ND SET` toggles `360` (30/360) or `ACT`. `CPT` at `DBD` gives the days.
- **Uneven cash flows.** `CF`, enter `CF0`, `C01`, `F01`, `C02`, and so on (`↓` moves between lines). Then `NPV`, enter `I` (percent), `↓`, `CPT` for the present value.

Chapter 2 and 3: Tax, Margin and Short Sales

2.1 Tax-equivalent yield

$$r_{TEY} = \frac{r_{muni}}{1-t} \qquad r_{AT} = r_{taxable}(1-t) \qquad t^* = 1 - \frac{r_{muni}}{r_{taxable}}$$

Where: r_{TEY} taxable yield equal to the muni after tax; r_{muni} municipal yield; $r_{taxable}$ taxable bond yield; r_{AT} after-tax yield; t marginal tax rate; t^* break-even tax rate (muni is better above it).

Calculator hint: plain arithmetic with `( )`; no TVM.

3.1 Margin purchase return

$$E = m_0 N P_0 \quad L = NP_0 - E \quad I = iL \quad r_{margin} = \frac{N(P_1 - P_0) + ND - I}{E}$$

Where: E own equity; m_0 initial margin; N shares; P_0 buy price; P_1 sell price; D dividend per share; L loan; i loan rate; I interest dollars.

Calculator hint: compute E, L, I in order and `STO` each; finish with one chain, then divide by `RCL` of E.

3.2 Margin call price

$$P_{call} = \frac{L/N}{1-m} \qquad L/N = P_0(1-m_0)$$

Where: P_{call} price that triggers the call; L/N loan per share (constant); m maintenance margin.

Calculator hint: plain arithmetic; the denominator $(1-m)$ needs parentheses.

3.3 Short sale return

$$E = m_0 N P_0 \qquad r_{short} = \frac{N(P_0 - P_1) - ND}{E}$$

Maximum gain $= NP_0$ (price falls to zero). Maximum loss is unlimited (price can rise without limit).

Where: N shares shorted; P_0 short-sale price; P_1 buy-back price; D dividend per share owed to the lender; m_0 initial margin; E margin posted.

Calculator hint: `STO` the margin E first, then divide the net profit by `RCL` E.

Chapter 4: Mutual Funds

4.1 Ending value after loads and expenses

$$V_T = V_0(1-\ell)(1+g-e)^T(1-c_T)$$

Set $\ell = 0$ for no front-end load and $c_T = 0$ for no back-end load (or after its schedule expires). Larger V_T means less fee drag.

Where: V_0 amount invested; ℓ front-end load; c_T back-end load charged at sale; g gross annual return; e expense ratio (include 12b-1 fees); T years held.

Calculator hint: TVM. Use $N = T$, $I/Y = (g - e)$ in %, PV = negative amount after any front-end load, $PMT = 0$, then `CPT FV`. Apply a back-end load afterwards by multiplying by $(1 - c_T)$.

Chapter 14: Bond Prices and Yields

14.1 Coupon interest. Annual interest $= F \times c \qquad C = \frac{F \times c}{n}$

Where: F face value; c annual coupon rate; n payments per year; C one coupon payment.

14.2 Free cash flow. $FCF = CFO - CapEx$

Where: CFO cash flow from operations; $CapEx$ capital expenditures.

14.3 Day count and accrued interest

$$AI = C \times \frac{d}{D} \qquad d_{30/360} = 360(Y_2 - Y_1) + 30(M_2 - M_1) + (D_2 - D_1)$$

30/360 (corporate, municipal): a 31st counts as the 30th; a semiannual period has $D = 180$. Actual/actual (Treasury notes and bonds): real days for d and D. Actual/360 (money market): real days, 360-day year.

Where: AI accrued interest; C one coupon payment; d days since last coupon; D days in the coupon period; Y, M, D year, month, day of the earlier (1) and later (2) dates.

Calculator hint: the `2ND DATE` worksheet counts days under `360` or `ACT`; then multiply C by d/D.

14.4 Bond price from a yield

$$P = \sum_{t=1}^{T} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^T} = C\,\frac{1-(1+y)^{-T}}{y} + \frac{F}{(1+y)^T}$$

Where: P price; C coupon per period; F face value; y yield per period; T periods to maturity.

Calculator hint: TVM with `CPT PV`; the sign of PV only shows direction.

14.5 Flat and full price. $P_{full} = P_{flat} + AI$

Where: P_{flat} quoted (clean) price; P_{full} invoice (dirty) price paid; AI from 14.3.

14.6 Yield to maturity

$$P = \sum_{t=1}^{T} \frac{CF_t}{(1+y)^t} \quad \text{(solve for } y\text{)}$$

Where: P price; CF_t cash flow in period t (coupon, plus F at the end); y YTM per period; T periods.

Calculator hint: TVM with `CPT I/Y`; enter the price as a negative PV. Check by entering I/Y and solving `CPT PV`.

Chapter 15: The Term Structure

15.1 Spot rates from a bill and a coupon bond

$$1+z_1=\frac{F}{P_{bill}} \quad y_1=2z_1 \qquad (1+z_2)^2=\frac{C+F}{P_{bond}-\dfrac{C}{1+z_1}} \quad y_2=2z_2$$

Where: P_{bill} 6-month bill price; P_{bond} 1-year coupon bond price; F face; C semiannual coupon; z_1, z_2 per-6-month spot rates; y_1, y_2 annual bond-equivalent spot rates.

Calculator hint: TVM gives each per-period spot rate: $N=1$ for the bill, $N=2$ for the bond's final payment after removing the first coupon's present value. Then double.

15.2 Forward rates and discount factors

$$f_n=\frac{(1+y_n)^n}{(1+y_{n-1})^{n-1}}-1 \quad f_1=y_1 \qquad DF_n=\frac{DF_{n-1}}{1+f_n}=\frac{1}{(1+y_n)^n}$$

Where: y_n n-year spot rate; f_n forward rate for year n; DF_n present value of \$1 received in year n.

Calculator hint: use `y^x`, `1/x` and `STO`/`RCL`; chain each discount factor from the one before.

15.3 Price from discount factors, then YTM

$$P=\sum_{t=1}^{T} CF_t \times DF_t$$

Where: CF_t cash flow in year t; DF_t discount factor from 15.2.

Calculator hint: multiply and add with `RCL` of the stored discount factors, `STO` the price, then use the YTM steps from 14.6.

Chapter 16: Bond Portfolios

16.1 Duration and price change

$$w_t=\frac{CF_t/(1+y)^t}{P} \quad D=\sum_{t=1}^{T} t\,w_t \quad D^*=\frac{D}{1+y} \quad \frac{\Delta P}{P}\approx -D^*\Delta y$$

A zero-coupon bond's duration equals its maturity.

Where: w_t weight of period t (PV of CF_t over price); D Macaulay duration (years); D^* modified duration; Δy yield change (decimal); $\Delta P/P$ percent price change.

Calculator hint: TVM (`CPT PV`) for each cash flow's present value, `STO` each, then one chain for P and one for $\sum t\,PV_t$.

16.2 Convexity

$$Cx=\frac{1}{P(1+y)^2}\sum_{t=1}^{T}\frac{CF_t}{(1+y)^t}(t^2+t) \qquad \frac{\Delta P}{P}\approx -D^*\Delta y+\tfrac{1}{2}Cx(\Delta y)^2$$

Where: Cx convexity; other symbols as in 16.1.

Calculator hint: reuse the stored present values; multiply each by (t^2+t), add, divide by $P(1+y)^2$.

16.3 Immunize a liability with two zero-coupon bonds

$$PV_L=\sum_t \frac{L_t}{(1+y)^t} \quad D_L=\sum_t t\,\frac{L_t/(1+y)^t}{PV_L}$$

$$w_1=\frac{T_2-D_L}{T_2-T_1} \quad w_2=1-w_1 \quad \text{Dollars}_j=w_j\,PV_L \quad \#\text{bonds}_j=\text{Dollars}_j(1+y)^{T_j}$$

Where: L_t payment due in year t; y market yield; PV_L present value of the liability (amount to invest); D_L liability duration; $T_1<T_2$ zero maturities; w_j portfolio weights; $\#\text{bonds}_j$ is the count of \$1-face zeros. A negative weight means selling that bond short.

Calculator hint: TVM for each liability's present value, `STO` them, then arithmetic for D_L and weights. The bond count uses TVM with `CPT FV`.

16.4 Cash-flow matching and why immunization is hard

$$\text{Cost today}=\sum_t \frac{L_t}{(1+y)^t}=PV_L$$

Buy a zero maturing on each payment date with face value L_t. No rebalancing and no yield-curve risk. Immunization needs rebalancing (duration drifts), protects only against small parallel shifts, and leaves convexity differences.

Calculator hint: TVM `CPT PV` for each payment, then add the sizes.

Additional Notes

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